



BilisOps Chat

Product Requirements Document — TikTok Shop App Review

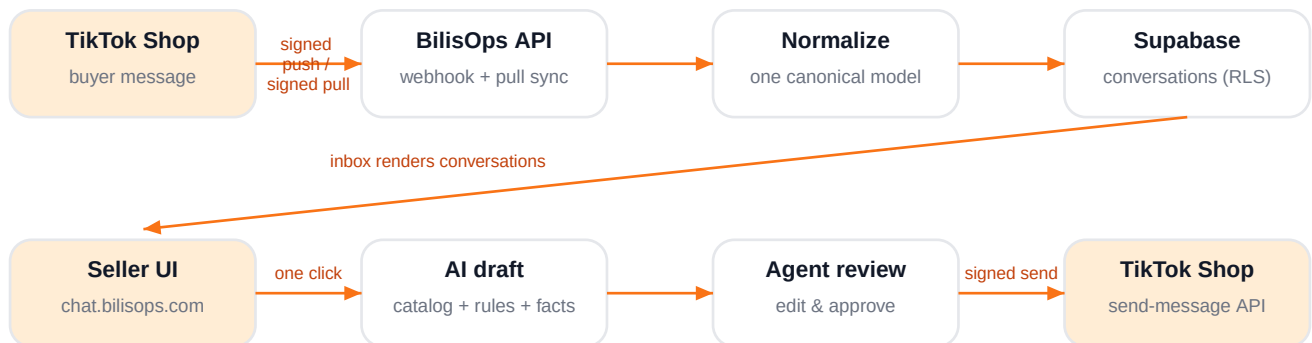
Category: Customer Service / Chat Management

BilisOps Chat is a customer-service workspace for TikTok Shop sellers. It brings a seller's buyer conversations into one inbox, helps agents reply faster with AI-assisted drafts grounded in the seller's own product catalog and policies, and reports service performance. This document describes the data flow, key features, and key use cases covered by the app.

Live product URL	https://chat.bilisops.com
Test account	tester@bilisops.com · password: admin123
Developer	BILISOPS SOFTWARE INSTALLATION SERVICE (DTI Business Name No. 8316712, Philippines)
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App key	6kj9do9stkf7l
Service scope	TikTok Shop sellers — Philippines & Southeast Asia (English / Tagalog / Taglish)

1. Architecture & data flow

The frontend and API run on Cloudflare (Pages + serverless functions); data is stored in Supabase (PostgreSQL) with per-seller isolation. All TikTok Shop calls are signed (HMAC-SHA256) and use the shop's OAuth access token.



2. Components

- **Frontend** — React SPA on Cloudflare Pages — inbox, AI controls, analytics dashboards
- **API** — Serverless functions (Cloudflare Workers runtime) — auth, OAuth callback, webhooks, chat sync, AI drafts
- **Database** — Supabase PostgreSQL — sellers, stores, conversations, orders; row isolation per seller
- **AI engine** — LLM drafts via server-side API — prompt is grounded in seller catalog, rules, and knowledge pack
- **TikTok integration** — OAuth authorization > token exchange > shop cipher; webhook receiver + per-minute pull sync; send-message API for replies

3. Key features

- **Official shop authorization (multi-shop)**

Sellers connect any number of TikTok shops through the official OAuth flow. Tokens are exchanged server-side and stored encrypted; shops can be removed anytime.

- **Unified chat inbox**

Buyer conversations from all connected shops in one workspace. New messages arrive via the message webhook and an automatic API pull sync (once per minute) with de-duplication by message id.

- **Reply to buyers**

Agent replies are delivered through the TikTok Shop customer-service send-message API. Delivery result is tracked per message.

- **AI-assisted drafts (human in the loop)**

One click drafts a suggested reply from: the buyer's last messages, the seller's Product Catalog, seller-written Reply Rules, and the shop's Knowledge Pack. The agent reviews and edits before sending — the AI never sends autonomously.

- **Product Catalog**

Seller-managed list of products (price, variants, stock, promo). The AI answers product questions only from this list; out-of-stock products trigger an in-stock alternative.

- **AI Reply Rules**

Seller-written rules enforced on every draft — e.g. polite po/opo forms, no discounts beyond listed promos, hand refunds to a human.

- **Per-shop Knowledge Pack**

Store facts (shipping, returns, policies) maintained per shop, with account-wide shared facts as fallback.

- **Sensitive-word filter**

Outgoing replies are checked against a seller-defined blocklist before send.

- **Analytics**

Response rate, average first-response time, resolution rate, and conversation-to-order conversion computed from chat + order events.

4. Key use cases

UC-1 · Connect a shop

Seller opens Settings > Store Authorization > selects TikTok > clicks "Connect TikTok store" > approves on TikTok's authorization page > redirected back; the shop appears as Authorized. Repeatable for multiple shops.

UC-2 · Receive a buyer message

Buyer messages the shop on TikTok > TikTok pushes the message webhook (signature verified) and/or the per-minute pull sync fetches it > the message is normalized and appears in the unified inbox with an unread badge.

UC-3 · Reply with AI assistance

Agent opens the conversation > clicks "AI Draft" > a suggested reply appears, grounded in catalog/rules/knowledge > agent edits if needed > clicks Send > reply is delivered via the TikTok send-message API.

UC-4 · Keep the AI accurate

Seller updates the Product Catalog (e.g. marks the hoodie Out of stock) and Reply Rules (e.g. "no discounts beyond listed promos") > from the next draft onward the AI reflects the change (suggests in-stock alternatives, declines discount requests politely).

UC-5 · Track service performance

Owner opens Analytics > Sales Conversion / Agent Performance > sees response rate, first-response time, and conversation-to-order conversion for the period, per store.

5. API scopes & why we need them

- **Shop authorized information** — After OAuth we read the authorized shop list (shop id, name, cipher) to route conversations to the right shop and sign subsequent calls.
- **Customer service — conversations & messages (read)** — Core of the product: syncing the shop's buyer conversations and message history into the unified inbox (webhook + pull).
- **Customer service — send message (write)** — Delivering the agent's approved reply back to the buyer inside TikTok Shop chat.

6. Endpoints registered with TikTok

OAuth redirect: <https://chat.bilisops.com/api/platforms/tiktok/oauth/callback>

Message webhook: <https://chat.bilisops.com/api/platforms/tiktok/webhook>

Webhook signature: HMAC-SHA256(app_secret, app_key + body), constant-time compared. Unsigned payloads are rejected in production mode.

7. Security & privacy

- **OAuth only** — We never ask for or store TikTok account passwords. Access is granted and revocable by the seller on TikTok.
- **Token safety** — Access/refresh tokens are stored server-side, never exposed to the browser, and all platform calls are signed server-side.
- **Tenant isolation** — Every record is keyed by seller id; one seller can never read another seller's conversations, shops, or analytics.
- **Data minimization** — We store only what the inbox needs: conversation messages, buyer display name/id, and order references for context.
- **Deletion** — Removing a shop (or revoking on TikTok) stops sync; synced data for that shop is deleted within 30 days on request.
- **Compliance** — Operated under the Philippine Data Privacy Act of 2012 (RA 10173) and TikTok Shop Partner policies.
- **AI boundary** — The AI only drafts; a human approves every outgoing message. Prompts contain only the seller's own catalog/rules/facts and the conversation being answered.



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